

Term 3 2026

FINAL EXAM ETL PROJECT

Business Intelligence
MADSC301-MUC07611

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01. INTRODUCTION

Analyze short-term trends in major U.S. semiconductor stocks

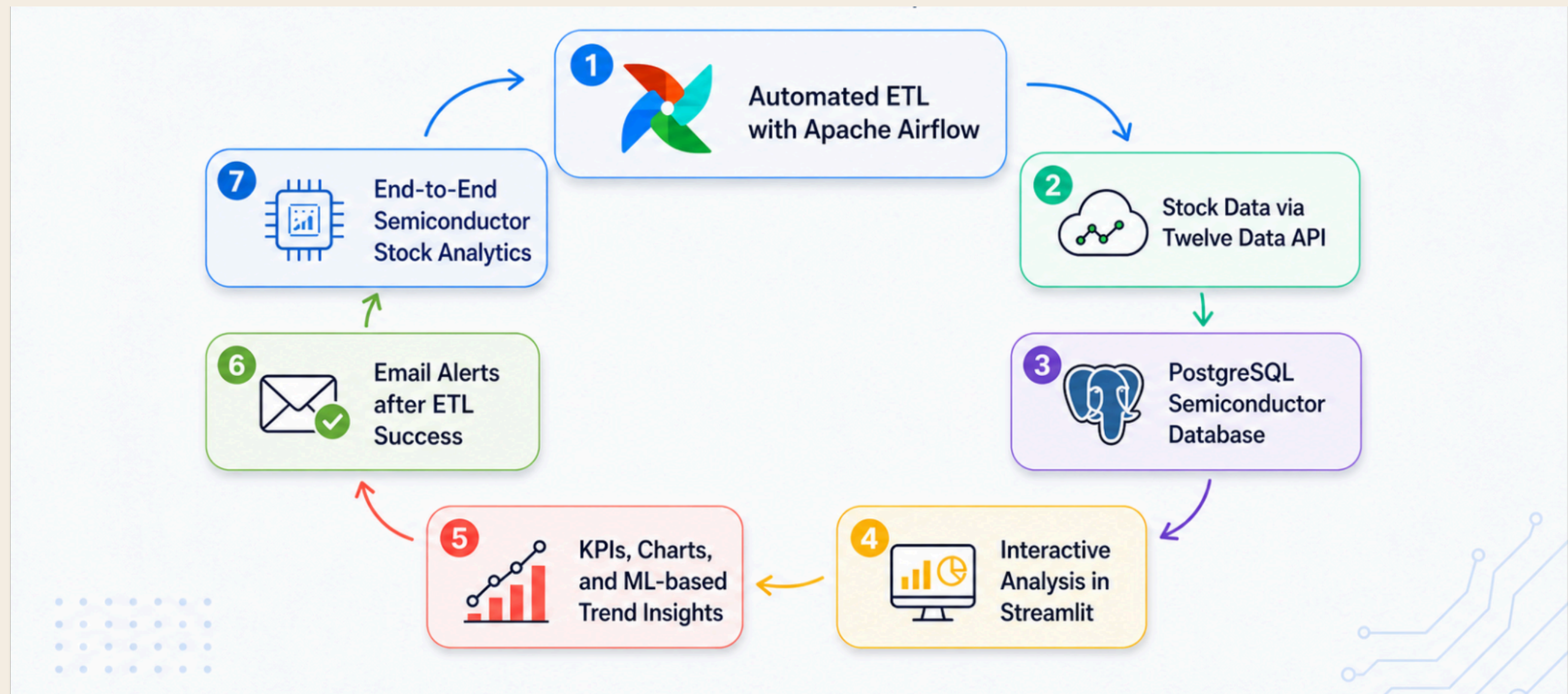
The project uses the **Twelve Data API**, **Apache Airflow**, **PostgreSQL**, and **Streamlit** to create an end-to-end workflow for semiconductor stock analysis.

The main objective of this project is to monitor and analyze short-term trends in major U.S. semiconductor stocks through an **automated ETL pipeline and interactive dashboard**. This project follows the assignment requirement to build a complete ETL pipeline including data collection, cleaning, storage, orchestration, and visualization.

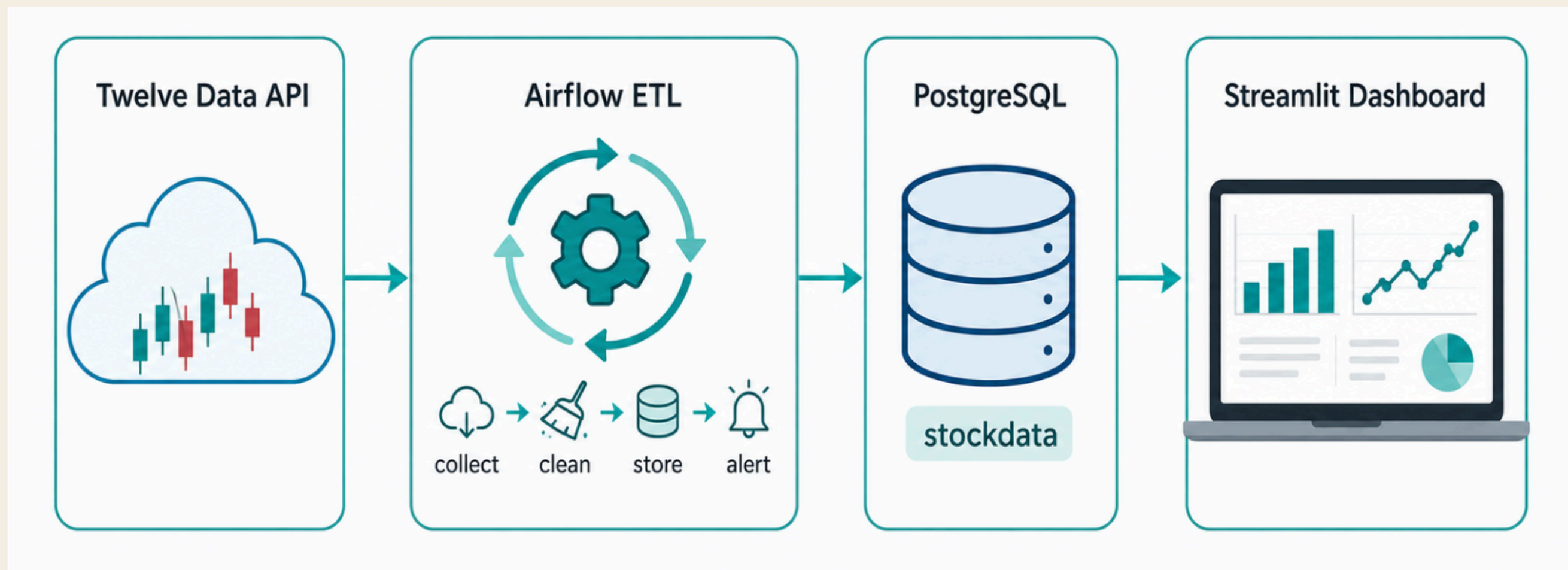


[image source](#)

02. KEY FEATURES



03. ETL ARCHITECTURE



04. ARIFLOW AUTOMATION

Day

airflow_etl

Run Dag

manual__2026-06-19T14:22:05.882880+00:00

00:03:32

00:01:46

collect

clean

store

alert

manual__2026-06-19T14:22:05.882880+00:00

success

Logical date

Execution type

Start date

End date

period

Triggered user name

2026-06-19 16:22:04

passivity

2026-06-19 16:22:06

2026-06-19 16:22:25

00:00:19.562

admin

Work instance

Asset Event

Audit Log

cord

Details

+ filter

Job ID

Map Index

situation

Start date

Number of attempts

Operator

period

alert

success

2026-06-19 16:22:23

1

PythonOperator

00:00:04.523

store

success

2026-06-19 16:22:18

1

PythonOperator

00:00:04.116

clean

success

2026-06-19 16:22:15

collect

success

2026-06-19 16:22:06

S

e@gmail.com

To: LEE Sujin

Fri 19/06/2026 16:26

The semiconductor stock data pipeline finished successfully.

Great news, thanks for the update.

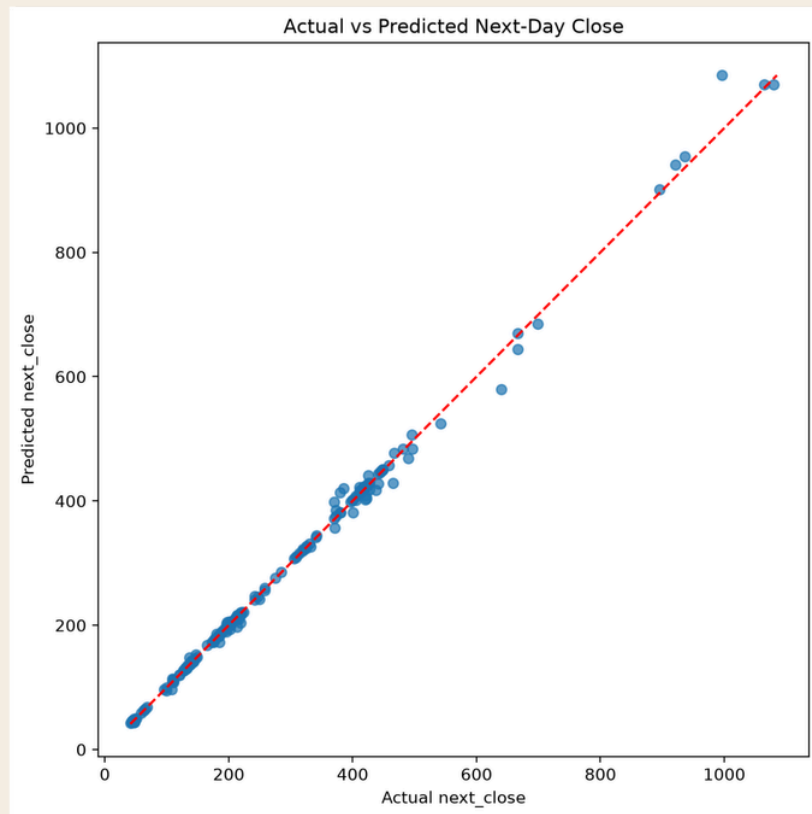
Great news! Thank you!

Good to hear. Thanks.

Reply

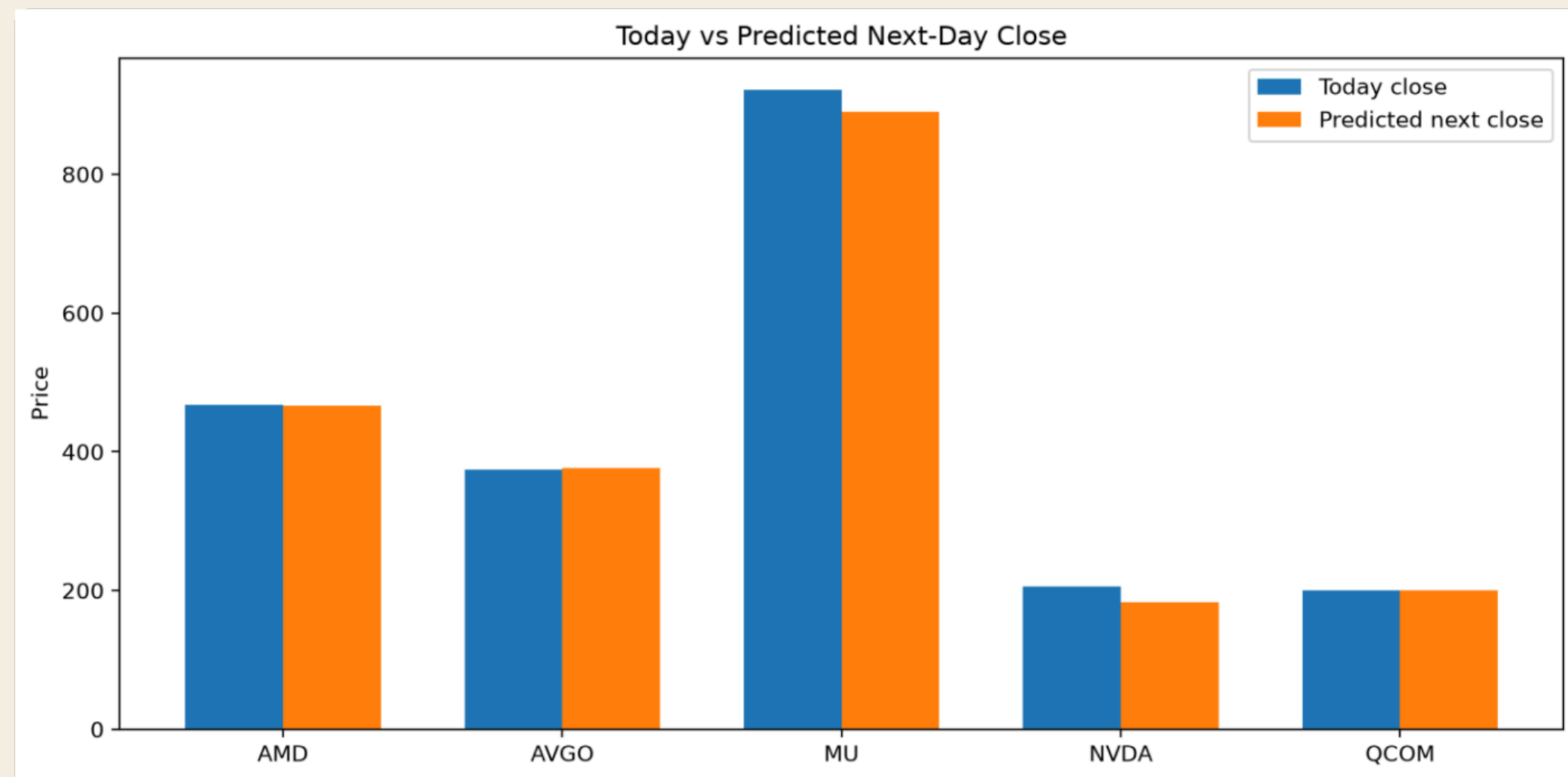
Forward

O5. MACHINE LEARNING

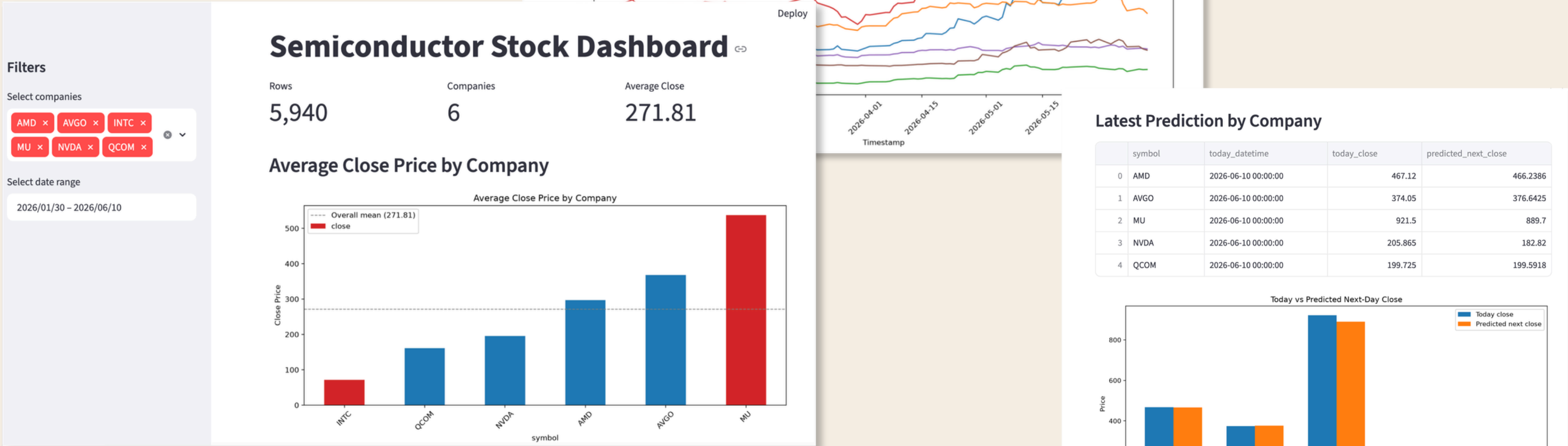


MSE 108.7496

R^2 0.9969



06. STREAMLIT DASHBOARD



O7. LINK

github: <https://github.com/leesujin113/Final-Term/tree/main>

08. AI USAGE

Images in slide 4,5 are created by AI

23.06.2026

THANK YOU